

Roll No.

MAB-201

B. TECH. (BIOTECH.) (SECOND SEMESTER)
END SEMESTER EXAMINATION, July/Aug., 2022
ENGINEERING MATHEMATICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) If $A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$ and $f(x) = 2x^2 - 3x + 5$, $g(x) = x^2 + 3x - 10$, then find $f(A)$ and $g(A)$. (CO1)

(b) Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$. (CO1)

(c) Solve the following system of equations by using matrix method : (CO1)

$$x + y + z = 6; x - y + z = 2, 2x + y - z = 1$$

P. T. O.

2. (a) Define continuity and check the continuity of the function : (CO2)

$$f(x) = \begin{cases} -x, & x < 0 \\ x, & 0 \leq x \leq 1 \\ 2-x, & x > 1 \end{cases}$$

- (b) Evaluate the following : (CO2)

(i) $\lim_{x \rightarrow 0} \frac{x}{\sqrt{2+x} - \sqrt{2-x}}$

(ii) $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{x^2}$

- (c) Find the maximum or minimum value of the function : (CO2)

$$y = 2x^3 - 9x^2 + 12x + 6.$$

3. (a) Define linear differential equation and solve : (CO3)

$$(x+1) \frac{dy}{dx} - y = e^x (x+1)^2$$

- (b) Solve : (CO3)

$$\frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} + 9y = 6e^{3x} + 7e^{-2x} - \log 2.$$

- (c) Solve the following differential equations : (CO3)

(i) $(D^2 - 5D + 6)y = e^x + e^{2x} + 5$

(ii) $(4D^2 + 12D + 9)y = 144e^{-3x}$

4. (a) Find a root of the equation $x^3 - x^2 - 1 = 0$ correct to two decimal places using bisection method. (CO4)

(b) Using Newton-Raphson method, find a root of the equation $x^4 - x = 10$ correct to three decimal places. (CO4)

(c) Evaluate $\int_0^1 \frac{dx}{1+x}$ considering five sub-intervals by : (CO4)

(i) Trapezoidal rule

(ii) Simpson's 1/3 rule

(iii) Simpson's 3/8 rule.

5. (a) Find the mean, median and mode for the following frequency distribution : (CO5, CO6)

Marks	No. of Students
10—20	9
20—30	10
30—40	18
40—50	16
50—60	14
60—70	15
70—80	12
80—90	6
90—100	2

- (b) Define correlation. Calculate Karl Pearson's coefficient of correlation from the following data : (CO5, CO6)

Marks in Statistics	Marks in Economics
84	78
51	36
91	98
60	25
68	75
62	82
86	90
58	62
53	65
47	39

- (c) Define Normal Distribution. In a radio manufacturing factory, average number of defective is 1 in 10 radios. Find out the probability of getting 2 defective radios in a random sample of 10 radios, using Poisson distribution. (CO5, CO6)

(Given that : $e^{-0.4} = 0.6703$.)

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Roll No.

TMA-201**B. TECH. (SECOND SEMESTER)****END SEMESTER EXAMINATION, July/Aug., 2022****ENGINEERING MATHEMATICS-II****Time : Three Hours****Maximum Marks : 100****Note :** (i) All questions are compulsory.(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Solve :

(CO1)

$$(D^2 + 2D + 1)y = e^{-x} \cos x + x; D \equiv \frac{d}{dx}.$$

(b) Solve :

(CO1)

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = \sin 2 \{ \log x \}$$

(c) Solve :

(CO1)

$$(D^4 - 3D^2 - 4)y = 5 \sin 2x - e^{-2x}; D \equiv \frac{d}{dx}.$$

2. (a) Write down the statement of Convolution theorem and hence to find the inverse Laplace transform of the function : (CO2)

$$f(s) = \left\{ \frac{1}{(s^2 + a^2)^2} \right\}.$$

P. T. O.

- (b) Define the Laplace transform and find the Laplace transform of the function : (CO2)

$$F(t) = te^{2t} \sin t.$$

- (c) Evaluate : (CO2)

$$L \left[e^{-4t} \frac{\sin 3t}{t} \right]$$

3. (a) Solve : (CO4)

$$\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} = \sin x \cos 2y$$

- (b) Solve : (CO4)

$$4 \frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 16 \log(x+2y)$$

- (c) Find a Fourier series expansion for the function $f(x) = x^2$ in the interval $(-\pi, \pi)$. (CO3)

4. (a) Using method of separation of variables to solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial y} + u(x, y);$

with the condition $u(x, 0) = 6e^{-3x}$. (CO5)

- (b) A string is stretched between the fixed points $(0, 0)$ and $(1, 0)$ and then released at rest from the position $u(x, 0) = A \sin \pi x$. Determine the displacement $u(x, t)$. (CO5)

- (c) Find the Fourier series for the function $f(x) = |x|$ in $-\pi < x < \pi$, Hence show that :

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8} \quad (\text{CO3})$$

(3)

5. (a) To Prove that :

(CO6)

$$(2n+1)xP_n(x) = (n+1)P_{n+1}(x) + nP_{n-1}(x).$$

(b) Prove that :

(CO6)

$$(i) J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

$$(ii) J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

(c) Find the temperature function $u(x, t)$ of one-dimensional heat equation

$$\frac{\partial u}{\partial t} = 3 \frac{\partial^2 u}{\partial x^2} \text{ with the boundary condition } u(0, t) = 0 = u(2, t) \text{ and the}$$

initial condition $u(x, 0) = x; 0 < x < 2.$

(CO5)

Roll No.

THU-201

**B. TECH. (SECOND SEMESTER)
END SEMESTER EXAMINATION, July/Aug., 2022**

ADVANCED PROFESSIONAL COMMUNICATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each sub-question carries 10 marks.

1. Answer any *two* of the following : (CO3)

(a) "Mastering the art of JAM Delivery." Explain in bullet points the ways to master it and explain each in 2-3 lines.

(b) Make a quick calculation of time of speaking and making an eye contact with fellow group members while speaking in a GD. Consider the number of people in the group as 10, discussion to last for 20 minutes and each member having to contribute for 4 times during the entire discussion.

P. T. O.

(c) Candidates were given 11:30 am as the time to report of the Interview. All the candidates were seated before time. The interviewer reached the venue at 11:30 am and while entering into the Interviewing Room, called the first candidate. The candidate entered the room and pulled a chair to sit. After taking the seat, he put his document file on the table and sat anticipating the first question/statement from the Interviewer. Interviewer gave him an intense look and then asked for his resume.

Simultaneously while reading the resume, the interviewer kept asking the questions as per the topics he was reading. Soon he got to know about the weak area of the respondent but continued to ask the questions from that area. Eventually the respondent panicked and failed to answer even the basic questions, said sorry and left.

Based on the above case, answer the questions below :

- (1) Comment on the attitude of the Interviewer. In addition, mention the styles of interviewing that the interviewer used.
- (2) Why do you think the Interviewer would have given the intense looks to the respondent ?
- (3) Asking questions from topics not known to the respondent is always treated as unethical and illegal. Comment.
- (4) Reading the last two sentences of the case, comment on the personality traits of the respondent.

2. Answer any *two* of the following :

(CO5)

- (a) Based on the below Job Application letter, answer the questions that follow the letter :

Rampur Road Haldwani, Uttarakhand, India

HR Manager

BMW Industries, India

07 July, 2022

Subject : To apply for the position of Powertrain Certification Engineer Crew (Job reference no. : 15439)

Sir/Ma' am :

Please accept this letter and the accompanying resume as an expression of my interest in the position of a Powertrain Certification Engineer Crew (Job reference no. : 15439). I have completed Bachelor of Technology degree in Mechanical Engineering from Graphic Era (deemed to be) University, Dehradun, Uttarakhand, India. I would like to begin employment with an organization in which my effective performance will be met with recognition and growth.

I have assimilated a comprehensive knowledge of leading edge engineering principles, tools, and practices, with emphasis on designing, building and testing of mechanical systems. I am proficient in the latest versions of Ansys, AutoCAD and SolidWorks. In addition, I have

P. T. O.

(4)

acquired hands-on exposure to engine assembly and cycle time calculation during my internships. I also have applied classroom learning to innovative and successful projects like Formula Student, SAE Supra and SHELL Eco Marathon in which I served as the Power-train lead and Technical Guide.

I am certain that I will be able to make significant contributions to your organization and would wholeheartedly welcome the opportunity to meet with you to discuss how my education and abilities may best be employed by your organization.

Devesh Manikaran

8059420000

deveshmani4@gmail.com

- (1) Comment on the correctness of format and placement of the date. If incorrect, provide the correction.
- (2) Why do you think the companies mention what to write as the subject?
- (3) Are the number of paragraphs used in writing this letter correct? Give the formula to calculate the number of paragraphs in any formal writing.
- (4) What do you think is incorrect in the sender's and recipient's address and in the closing of the letter? Provide necessary corrections.

- (b) Tesla has announced vacancies for the position of a Graduate Engineer Trainee with an experience of 0–2 years. You have completed B. Tech in CS in 2022 and you had a gap of I year in between XII and B. Tech.

Based on the above information, answer the following :

1. Write the Career Objective for the Resume to apply for the mentioned vacancy.
 2. Because the experience is mentioned as 0–2 years, should we be writing the academic detail and employment detail both or either. Comment.
 3. What style of writing the resume should the candidate opt for ?
 4. Write a paragraph of the cover letter (only para, not the entire letter) describing your candidature for the position.
- (c) While presenting a critical review of the movie “K.G.F. Chapter 2”, the presenter either read from the sheets he was carrying or tried to recall from memory whatever he had mugged up. Finding it difficult to adjudge the candidate’s potential, the panelists asked him to present the reviews of the movie “Samrath Prithviraj”. Though he spoke on the movie but used a lot of segregators and fillers in between and finished it by fits and starts.

Based on the situation, answer the following questions :

- (1) Comment on how successful the person was in building a rapport with the panelists and the audience ? Given reason for your stance.
- (2) Comment on the speaking style.

(3) Give corrective measures using which the presenter would have been effective.

(4) The speaker presented the review of "Samrath Prithviraj" using which style of speech delivery?

3. Answer any *two* of the following :

(CO4)

(a) While going through the proposal of a project, the reader was not convinced with the practical viability of the project. Also the budget demand sounded very big to him.

The project was about making a device to support the mal-functioning heart of human body taking the power from the body signals. And the budget proposed was 2 crore INR. He thought that taking the signals from human body and then amplifying those signals was not very sensible as would involve a lot of noise. In addition, he thought the cost proposed was way too big for this. Therefore, he suggested that the project should either be dropped or be altered as per his findings and recommendations.

Based on the case, answer the following questions :

1. Suggest which report should he be writing to state his findings on the project?
2. While writing the report, the writer used very simple vocabulary. Comment on the report's quality in terms of a technical report.

3. The writer though gave the recommendations but did not append any data or information that the reader may want to go through with. What went missing that caused this report to be ineffective ?
4. In maximum three lines write to the proposer of the project about the rejection of the project.

(b) Write short notes on Plagiarism and Objectives of Research.

(c) Write short notes on Letter of Transmittal and Draft Contract.

4. Answer any *two* of the following : (CO1, CO2)

(a) Explain all 5 ways of developing a paragraph while writing an essay.

(b) Provide :

(1) Two linkers/transition words for expressing **Importance**

(2) Two linkers/transition words for showing **Similarity**

(3) Two linkers/transition words showing **Difference**

(4) Two linkers/transition words showing **Cause and Effect**

(5) Two linkers/transition words showing **Time**

(c) Write any 5 rules to write a Precis.

5. Answer any *two* of the following : (CO1, CO2)

(a) Provide :

(1) Phrasal verb expressing **Inventing a story**.

(2) Idiom expressing the meaning of the phrase underlined : I enjoyed my tour to Paris more because as a welcome gesture it was **on credit** from the company that I had recently joined.

(3) Oxymoron expressing **Weak enthusiasm**.

P. T. O.

(4) Phrasal verb expressing the same meaning as the phrase underlined : The project that I had proposed has been **rejected**.

(5) Idiom expressing the meaning **Totally**

(b) Provide :

(1) Oxymoron meaning **"Angry but not showing it willingly"**

(2) Phrasal Verb expressing the meaning **Resemble**

(3) Idiom meaning **Honest**

(4) Oxymoron meaning **Complete absence of sound**

(5) Phrasal Verb meaning **Shop around**

(c) Provide :

(1) Idiom meaning **Irregularly**

(2) Oxymoron meaning **Something that serves no purpose.**

(3) Phrasal Verb meaning **Become popular**

(4) Idiom meaning **Costly or Troublesome person**

(5) Oxymoron meaning **False idea**

Roll No.

TCS-201

B. TECH. (SECOND SEMESTER)

END SEMESTER EXAMINATION, July/Aug., 2022

PROGRAMMING FOR PROBLEM SOLVING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Kholi likes the number 239. Therefore, he considers a number *lucky* if its last digit is 2, 3, or 9.

Kholi wants to watch the numbers between L and R (both inclusive), so he asked you to determine how many *lucky* numbers are in this range ?

Can you help him ?

(CO1, CO3)

P. T. O.

(2)

(b) (i) #include<stdio.h>

#include<string.h>

int main()

{ struct box

{ char shape[10];

} box1, box2;

strcpy(box1.shape, "TRIANGLE");

printf("%s", box1.shape);

box2 = box1;

strcpy(box2.shape, "SQUARE");

printf("%s", box1.shape);

return 0;}

(ii) #include<stdio.h>

#include<string.h>

int main()

{ struct box

{ char shape[10];

} box1, box2;

strcpy(box1.shape, "TRIANGLE");

printf("%s", box1.shape);

box2 = box1;

strcpy(box2.shape, "SQUARE");

printf("%s", box1.shape);

return 0;}

(iii) #include<stdio.h>

```
int main( )  
{ char str[ ] = "OUR UNIVERSITY";  
  char *s = str;  
  printf("%s\n",s++ +5);  
  return 0;  
}
```

(iv) list 1=["blue","green",3,7, "red"]

```
list1.append("85")  
list1.extend([3,"black","white"])  
print(list1)  
list1.pop(2)  
list1.pop( )  
print(list1)
```

(v) #include<stdio.h>

```
void chane(int*,int);  
int main( )  
{  
  int i, a[ ] = {2, 4, 6, 8, 10};  
  change(a, 5);  
  for(i=0; i<=4;i++)  
    printf("%d,",a[i]);  
  return 0;  
}
```

```
void change(int*b,int n)
```

```
{
```

```
    int i;
```

```
    for(i=0; i<n; i++)
```

```
        *(b+1)=*(b+i)+5;
```

```
}
```

(CO1, CO3)

(c) (i) `#include<stdio.h>`

```
int main( )
```

```
{    int arr[3] = {2, 3, 4};
```

```
    char *p;
```

```
    p = arr;
```

```
    p = (char*)((int*)(p));
```

```
    printf("%d",*p);
```

```
    p = (int*)(P+1);
```

```
    printf ("%d",*p);
```

```
    return 0;}
```

(ii) `#include<stdio.h>`

```
#include<string.h>
```

```
int main( )
```

```
{    char *str;
```

```
    str = "%S";
```

```
    printf(str, "K\n");
```

```
    return 0;}
```


(iii) #include<stdio.h>

```
int main( ) { int a, b, c;
    char *p = 0;
    int *q = 0;
    double *r = 0;
    a = (int)(p + 1);
    b = (int)(q + 1);
    c = (int)(r + 1);
    printf("%d %d %d",a,b,c);
    return 0;}
```

(iv) a = input("enter a number") #assume user input 5

b = input("enter another number") #assume user input 10

c = a+b

print(c)

print(a+b)

(v) #include<stdio.h>

void change (int*, int);

int main()

```
{ int i, a[ ] = {2, 4, 6, 8, 10};
```

```
    change(a,5)
```

```
    for(i=0; i<=4; i++)
```

```
        printf("%d, ",a[i]);
```

```
    return 0;}
```

void change (int *b, int n)

```
{ int i;
```

```
    for(i=0; i<n; i++)
```

```
        *(b+i) = *(b+i+2)+3;}
```

(CO1, CO3)

P. T. O.

2. (a) Explain pointers using a example. Also, write code to how dynamic memory allocation functions helps in saving memory. (CO1, CO3)
- (b) Draw a flowchart to input an array and sort it using pointer. (CO1, CO3)
- (c) Write a C code to input a string and count how many characters have single and multi-occurrence in the input string. (CO1, CO3)
3. (a) Explain any *four* from the following : (CO2, CO4)
- (i) Nesting of structures
 - (ii) Self-referential structure
 - (iii) typedef in structure
 - (iv) Comparisons and copy operation between two structure variables
 - (v) Union and bitfield
- (b) Draw a flowchart to calculate gross salary and net salary of n employees working in a retail medical shop. If their basic DA, TA and other allowances, deductions are given display the employee name, employee ID, month/year of salary whose net salary is greater than ₹ 50,000.
- Gross Salary = Basic pay + All other allowances
- Net Salary = Gross Salary – Deduction (CO2, CO4)
- (c) Write a C code to create a structure student with data members roll_no, name and marks in 3 subjects. Input details of 10 students and print the detail of student having maximum total marks. (CO2, CO4)
4. (a) What is the need of file in C ? Differentiate between text and binary file. Explain with syntax any *three* sequential access and *two* random access functions in file handling. (CO2, CO5)

(b) Write an algorithm to read n integer in a file and copy sum of digit of each number in another file. (CO2, CO5)

(c) Write a C code to input a string in a file and print it in reverse order. (without using fgets()) (CO2, CO5)

Example :

File content : Hi how are you ?

Output : ?uoy era who iH

5. (a) Explain any *five* features of Python which are different from C language. What are the different application areas of Python ?

(CO2, CO6)

(b) Write a Python code to input five numbers and print the largest among them. (CO2, CO6)

(c) Write a Python code to print the sum of even numbers between given range P and Q : (CO2, CO6)

Input : P = 5 and Q = 15

Output : $6 + 8 + 10 + 12 + 14 = 50$

Roll No.

BTE-201

B. TECH (SECOND SEMESTER)

END SEMESTER EXAMINATION, July/Aug., 2022

ADVANCED ORGANIC CHEMISTRY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Comment on the cyclic structure of glucose; draw appropriate diagrams.

Draw the Haworth projection formula of glucose and discuss. (CO5)

(b) Differentiate between Kilyani Fischer and Ruff degradation in glucose.

(CO5)

(c) Classify carbohydrates with suitable examples. Give *five* functions of carbohydrates.

(CO5)

P. T. O.

2. (a) What is the role of buffer ? Differentiate between acidic and basic buffer. What is the pH of the solution containing 0.1 mole of ephedrine and 0.01 mole of ephedrine hydrochloride per liter of solution. pK_b of ephedrine is 4.64. (CO4)
- (b) Write the basics of Thermogravimetric analysis and its role in analytical chemistry. (CO4)
- (c) What type of information is obtained by studying the FTIR and UV Spectra of an organic compound. (CO4)
3. (a) Give the principle of NMR spectroscopy. Predict the number of signals and splitting pattern in the NMR spectrum of : (CO4)
- (i) $CH_3 - CH_2 - CH_2 - OH$
- (ii) $CH_2 - OH - CH_2 - OH$
- (iii) $CH_3 - CH_2 - CH_2 - CH_3$
- (b) Enumerate the working principle of adsorption chromatography and partition chromatography.
- What volume of 0.08 M HCl will be required to neutralize 25 cc of 0.02 M Na_2CO_3 solution ? (CO4)
- (c) Give list of important electrophiles, their source and the end product of electrophilic aromatic substitution reaction in benzene. Give the mechanism of halogenations of benzene. (CO2)
4. (a) Enumerate the properties and some applications of nanomaterials. (CO3)
- (b) Discuss how azeotropic distillation can help in the purification of organic compounds. (CO1)

(3)

- (c) Use of catalysts is one of the principles of green chemistry. Discuss with examples. (CO3)
5. (a) What reaction products are expected when tert-butyl chloride is stirred in a mixture of ethanol and water ? (CO2)
- (b) How do substituent groups on an aromatic ring influence the course of electrophilic aromatic substitution ? Classify them by their effects. (CO2)
- (c) Single bonds adjacent to double and triple bonds are progressively shorter and stronger than single bonds adjacent to other single bonds : comment and discuss with examples. (CO2)

Roll No.

TEV-201

**B. TECH. (SECOND SEMESTER)
END SEMESTER EXAMINATION, July/Aug., 2022**

ENVIRONMENTAL SCIENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Environment. Discuss components of Environment in detail.
(CO1)
- (b) Define Ecological Pyramids. Discuss various types of Ecological Pyramids.
(CO1)
- (c) Explain the concept of energy and nutrient flow in an ecosystem. (CO1)
2. (a) What is meant by Mineral Resources ? Explain the adverse effects of mining on environment.
(CO1)
- (b) What is conservation of biodiversity ? Discuss methods of conservation of biodiversity.
(CO1, CO3, CO4)

P. T. O.

- (c) What is meant by alternate energy sources ? How it will be beneficial for environment and mankind ? (CO1, CO3, CO4)
3. (a) Discuss waste water treatment plant in detail. (CO1, CO5)
- (b) Define soil pollution. Discuss its effects and methods to reduce soil pollution. (CO1, CO5)
- (c) Describe the processes adopted for solid waste disposal with their advantages and disadvantages. (CO1, CO5)
4. (a) Write short notes on the following : (CO1, CO2)
- (i) Chipko movement
- (ii) Consequences of deforestation
- (b) Define Thermal Pollution. Discuss various ways of reducing thermal pollution. (CO1, CO2)
- (c) Define Bio-geochemical cycle. Discuss Nitrogen Cycle in detail. (CO1, CO2)
5. (a) How sustainable development is the key to conserve environment ? Explain with appropriate example. (CO3)
- (b) Define disaster. What methods would you suggest to reduce the impact of landslides ? (CO3)
- (c) Write short notes on the following :
- (i) Ozone Layer Depletion
- (ii) Wildlife Protection Act, 1972

Roll No.

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End Semester Examination 2022

(Set – A)

Name of the Course: B. Tech

Semester: 2ND

Name of the Paper – Basic Electrical Engineering

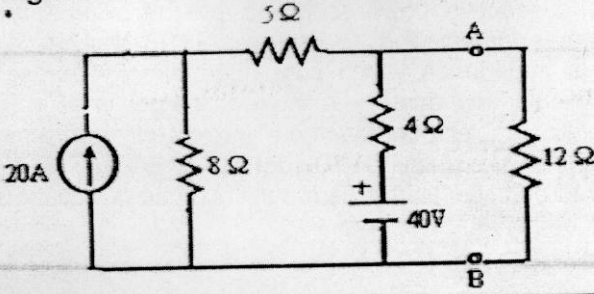
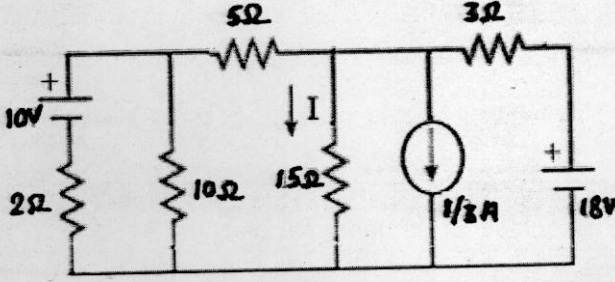
Paper Code: TEE 201

Time: 3 Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)
(a)	Give the Statement of Norton's Theorem. Find the current in 12-ohm resistance using Norton theorem for the given circuit. 
(b)	Determine current through 15-ohm resistance by nodal analysis. 
(c)	Derive an expression of delta to star transformation with example.
Q2	(20 marks)
(a)	Explain Resonance. Derive the expression for resonating frequency in Series Resonance. A coil of resistance 200 Ω and inductance 200 μH is connected in series with a 100-pF capacitor. The circuit is connected to a 15 V variable frequency source. Calculate: (i) the resonant frequency (ii) current at resonance (iii) voltage across L and C at resonance.
(b)	What are the line and phase quantities in 3 φ AC System? Derive the relation between Line and Phase quantities for balanced and symmetrical 3 φ star connected system.
(c)	Calculate the average value, r.m.s value, form factor and peak factor of ac voltage whose waveform is shown in Figure below

Q3	(20 marks)	
(a)	An Ideal transformer has a turn ratio 100/400. The LV winding is connected to a source of 3.3kV, 50Hz. An impedance of $(90+j30) \Omega$ is connected across the secondary terminals. Calculate: - The value of maximum core flux, - The primary and secondary currents, - Real and reactive power supplied by the source to the transformer primary, and - Value of impedance which connected directly across the source would draw same real and reactive power as in (iii)	
(b)	Figure 1 shows a rectangular magnetic core with an air gap. Find the exciting current needed to establish a flux density of $B_g=1.3T$ in the air gap, Given $N=500$ turns and Relative permeability of the core $\mu_r=5000$.	
(c)	Discuss various types of losses occurring in the transformer and how are these losses are accounted for efficiency calculation.	
Q4	(20 marks)	
(a)	Derive EMF equation of a DC generator. What will be change in EMF induced if flux is reduced by 20% and the speed is increased by 20%.	
(b)	Explain the working principle of 3-phase induction motor with suitable diagram. Also explain the importance of slip.	
(c)	Explain principle of electromechanical energy conversion in motor and generator. Enumerate all the parts of a DC machine.	
Q5	(20 marks)	
(a)	What do you understand by switchgear? Give classification of low voltage (LV) switchgear. Compare the characteristics of MCB and MCCB.	
(b)	How power factor can be defined? Write down the causes of low power factor. What are different techniques of power factor improvements, explain any one in detail.	
(c)	What are the basic elements of a battery cell? Give classification of batteries. Write down the characteristics of Lead-acid batteries.	

Roll No.

TEC-201

B. TECH. (SECOND SEMESTER)

END SEMESTER EXAMINATION, July/Aug., 2022

BASIC ELECTRONICS ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Solve the following :

(CO1)

(i) $(758.35)_9 + (3562.45)_7 = ()_6$

(ii) $(6243.44)_7 - (3665.32)_7 = ()_9 = ()_4$

(iii) $(111000111.11111)_2 \div (111)_2 = ()_2$

(b) (i) Prove that :

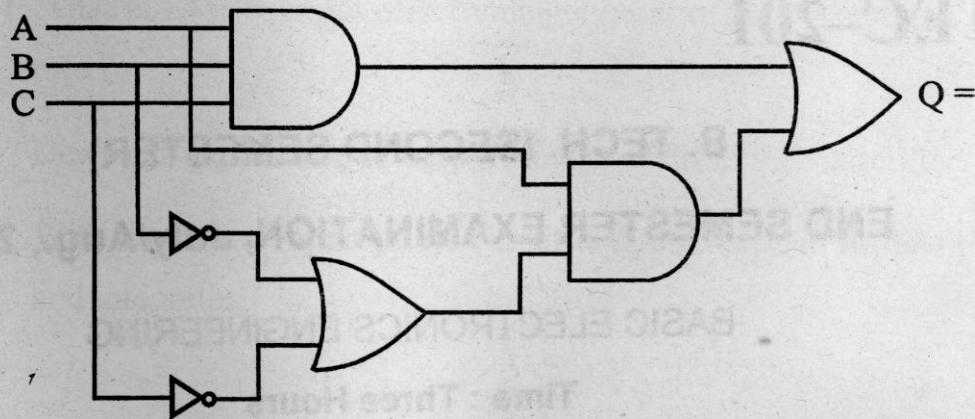
$$A'B = BC' + BC + AB'C' = B + AC'$$

(ii) Implement EX-OR as well as EX-NOR using NAND only and NOR only.

(CO1)

P. T. O.

- (iii) Find the expression for output and draw the truth table for the circuit given below : (CO1)



- (c) (i) Convert the following equation to standard form and solve it with the help of K-map :

$$F(A, B, C, D) = B'D + A'D + BD$$

- (ii) Solve the following equation with the help of K-map and implement it with NOR only : (CO1)

$$F(A, B, C, D) \Sigma m(2, 3, 6, 7, 8, 10, 13, 14, 15) + \Sigma d(0, 1, 5, 12)$$

2. (a) (i) What is Zener diode ? Explain its operating principle with the help of its I-V characteristics.
- (ii) The voltage across silicon diode at room temperature is 0.71 V when 2.5 mA current flows through it. If the voltage increases to 0.8 V, then calculate the new diode current. (CO2)
- (b) (i) Explain the formation of depletion layer in PN junction diode in forward bias condition with its I-V characteristics.

- (ii) A silicon bar is 0.2 mm long and has a cross-sectional area 0.04 mm^2 . A voltage source of 1 V is applied across the bar which results in a current of 8 mA. Assuming that the current is due to electrons and electron mobility is $0.36 \text{ m}^2/\text{V-sec}$. Then calculate :

(CO2)

- (A) concentration of free electrons.
 (B) Drift velocity.
- (c) (i) Derive an expression for drift and diffusion current in semiconductor.
- (ii) In a germanium sample, a donor type impurity is added to the extent of 1 atm. per 10^8 germanium. The density of Ge atoms is $4.41 \times 10^{22} / \text{cm}^3$. Calculate the resistivity of germanium. (CO2)
- Given :

$$\mu_n = 3800 \text{ cm}^2 / \text{V-sec}$$

$$\mu_p = 1800 \text{ cm}^2 / \text{V-sec}$$

$$n_i = 2.5 \times 10^{13} / \text{cm}^3$$

3. (a) (i) Derive an expression for average and r.m.s. current for half wave rectifier.
- (ii) In the bridge rectifier circuit, the diodes are assumed to be ideal. Find :
- (A) DC output voltage
 (B) Peak inverse voltage
 (C) Output frequency

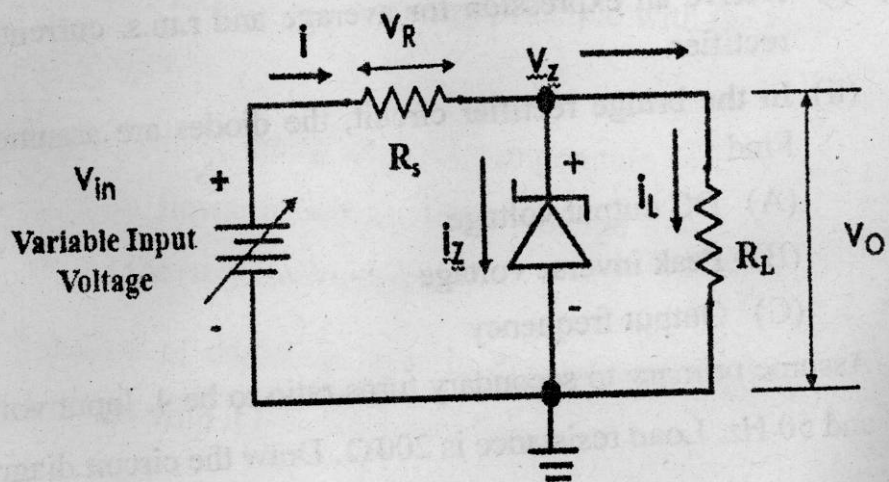
Assume primary to secondary turns ratio to be 4. Input voltage is 230 V and 50 Hz. Load resistance is 200Ω . Draw the circuit diagram as well.

(CO3)

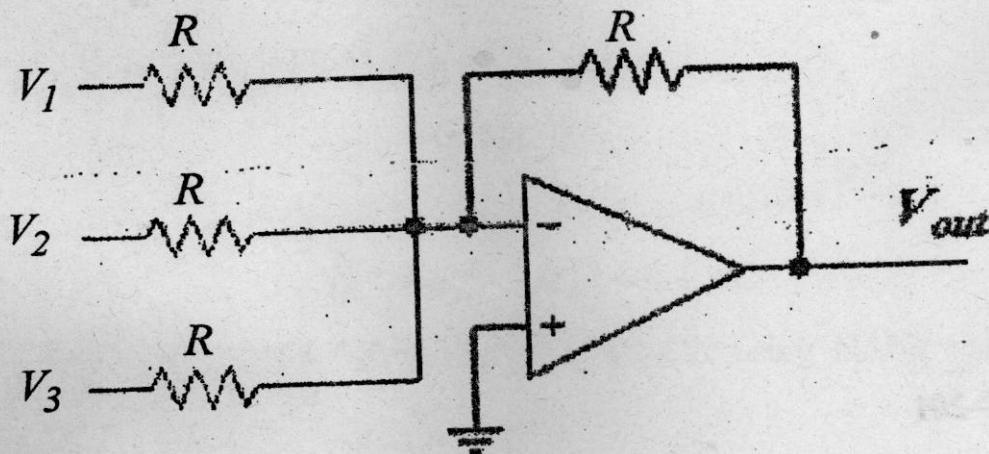
P. T. O.

(b) A crystal diode having internal resistance 20Ω is used for half-wave rectification. If the applied voltage is $50 \sin \omega t$ and load resistance is 800Ω . Then find and draw the circuit diagram : (CO3)

- (i) I_m, I_{dc}, I_{rms}
 - (ii) a.c. power input
 - (iii) d.c. power output
 - (iv) d.c. output voltage
 - (v) efficiency of rectification
- (c) (i) What is transformer utilization factor ? Derive an expression for transformer utilization factor for half wave and full wave (bridge and center tap) rectifier.
- (ii) Explain the working of voltage regulator. Calculate I_S or I, I_L, I_Z and P_Z for the circuit given below if : $V_{in} = 70$ and 110 V, $R_S = 5k\Omega$, $R_L = 10k\Omega$ and $V_Z = 50$ V. Show that the load current doesn't change even though the diode current changes for $V_{in} = 70$ and 110 V. (CO3)



4. (a) (i) Draw the common collector configuration circuit and explain its current gain, input and output characteristics.
- (ii) In a transistor if $\alpha = 0.995$ and emitter current is 10 mA, I_{CBO} is $0.5 \mu A$, then determine I_C , I_B , β and I_{CEO} . (CO4)
- (b) (i) Derive d.c. load line, zero signal base current, operating point (Q) and stability factor for the emitter bias circuit.
- (ii) For a fixed bias circuit, draw d.c. load line and locate Q point if the transistor used is made of Si with $\beta = 100$, $V_{CC} = 9 V$, $R_B = 830 k\Omega$ and $R_C = 3 k\Omega$. (CO4)
- (c) (i) Explain the self-bias (voltage divider) circuit and derive its operating point (Q) coordinates.
- (ii) Derive the relationship between α , β and γ . (CO4)
5. (a) (i) Derive the block diagram of op-amp. and derive an expression for output voltage of summing amplifier.
- (ii) For the circuit shown in figure, determine output voltage if $V_1 = V_2 = V_3 = 20 V$: (CO5)



P. T. O.

- (b) (i) Derive an expression for output voltage of inverting and non-inverting op-amp.
- (ii) What is CMRR, input offset voltage, input offset current and slew rate ? (CO5)
- (c) (i) Derive an expression for output voltage of difference amplifier.
- (ii) With the help of appropriate circuit diagram, explain differentiator and integrator. (CO5)

Roll No.

TPH-201

B. Tech. (SECOND SEMESTER)

END SEMESTER EXAMINATION, July/Aug., 2022

ENGINEERING PHYSICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Derive the expression for diameter of bright rings in Newton's ring arrangement having refractive index μ between the lens and the plate.

(CO1)

(b) Deduce the expression for intensity in N-slit Fraunhofer diffraction and deduce the position of principal maxima.

(CO1)

(c) A single slit is illuminated by light composed of two wavelengths λ_1 and λ_2 . One observes that the due to Fraunhofer diffraction the first minima obtained for λ_1 coincides with the second diffraction minima of λ_2 . What is the relation between λ_1 and λ_2 .

(CO1)

P. T. O.

2. (a) Discuss the method for production of plane, elliptically and circularly polarized light. (CO2)
- (b) Explain the construction and working of Helium Neon Laser along with the energy level diagram. (CO2)
- (c) A 5% solution of cane sugar placed in tube of length 40 cm, causes the optical rotation of 20° . How much length of 10% solution of the same substance will cause 35° rotation ? (CO2)
3. (a) Discuss the Einstein's postulates of special theory of relativity and derive the expression for length contraction. (CO4)
- (b) Explain frame of reference and derive Einstein mass energy relation. (CO4)
- (c) The mass of a moving electron is 11 times its rest mass. Calculate its kinetic energy and momentum. [Rest mass of electron = 9.1×10^{-31} kg] (CO4)
4. (a) Write short notes on Type I and Type II superconductors. (CO3)
- (b) Write Maxwell's equations in integral form and derive first three Maxwell's equations. (CO3)
- (c) Discuss Meissner effect in superconductor. (CO3)
5. (a) Derive eigenvalue and eigenfunction for particle confined in one dimensional infinite potential box. (CO5)

(3)

(b) Define de-Broglie wavelength and explain the Heisenberg uncertainty principle. (CO5)

(c) An electron microscope is used to locate a particle to locate an electron in an atom within a distance of $0.2\text{\AA}$. What is the uncertainty in the momentum of the particle located in this dimension? (CO5)

Roll No.

TCH-201

B. TECH. (SECOND SEMESTER)

END SEMESTER EXAMINATION, July/Aug., 2022

ENGINEERING CHEMISTRY

Time : Three Hours

Maximum Marks : 100

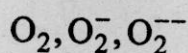
Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) With the help of molecular orbital diagram, arrange the following molecules in order of their increasing bond length : (CO1)



- (b) What do you mean by intermolecular and intramolecular hydrogen bonding ? Explain the following : (CO1)

- (i) Why glycerol is more viscous than ethanol ?
(ii) Why *p*-nitrophenol is more soluble than *o*-nitrophenol in water ?
(iii) Why ice floats over water?

P. T. O.

Roll No.

TCH-201

B. TECH. (SECOND SEMESTER)

END SEMESTER EXAMINATION, July/Aug., 2022

ENGINEERING CHEMISTRY

Time : Three Hours

Maximum Marks : 100

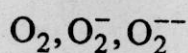
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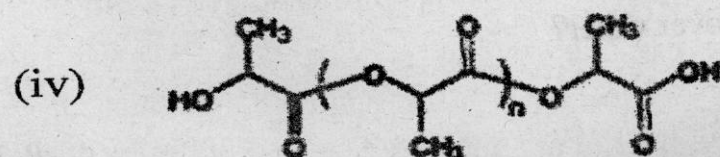
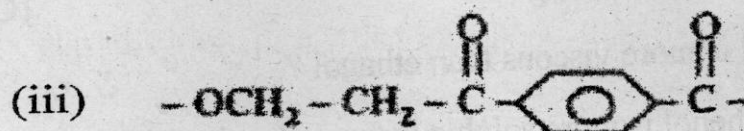
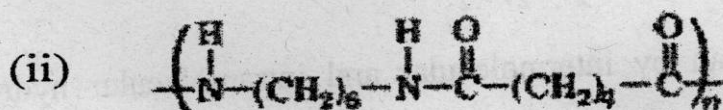
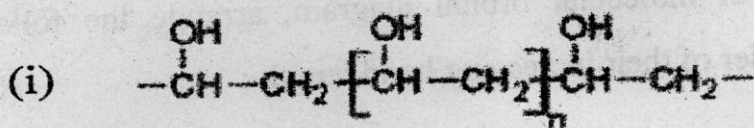


- (b) What do you mean by intermolecular and intramolecular hydrogen bonding ? Explain the following : (CO1)

- (i) Why glycerol is more viscous than ethanol ?
- (ii) Why *p*-nitrophenol is more soluble than *o*-nitrophenol in water ?
- (iii) Why ice floats over water?

P. T. O.

- (c) What do you mean by nanostructured materials ? Discuss their classification, properties and applications. (CO1)
2. (a) Discuss the zeolite process for the removal of hardness of water. Write its merits over soda-lime process. (CO2)
- (b) What do you mean by temporary and permanent hardness ? Calculate the amount of lime (85% Purity) and soda (90% Purity) required to soften 25000 L of water having the following analysis : (CO2)
- $\text{Ca}(\text{HCO}_3)_2 = 4.86 \text{ ppm}; \text{Mg}(\text{HCO}_3)_2 = 7.3 \text{ ppm}; \text{CaSO}_4 = 6.8 \text{ ppm};$
 $\text{MgCl}_2 = 5.7 \text{ ppm}; \text{MgSO}_4 = 9.0 \text{ ppm}; \text{NaCl} = 5.85 \text{ ppm}$
- (c) Write short notes on the following : (CO2)
- (i) Calgon conditioning
 - (ii) Scale and sludge
 - (iii) Priming and foaming
 - (iv) Caustic embrittlement
3. (a) What do you mean by functionality and degree of polymerization ? Identify name of the polymer in the following and draw the structure of monomers : (CO3)



- (b) Explain mechanisms of free radical and cationic polymerizations. (CO3)
- (c) (i) Differentiate between thermosetting and thermoplastics by giving suitable examples.
- (ii) Write a note on liquid crystalline polymers. (CO3)
4. (a) Define GCV and NCV. On burning 0.83 g of solid fuel in a bomb calorimeter, the temperature of 3500 g of water rises from 26.5°C to 29.2°C . Water equivalent of calorimeter and latent heat of steam are 385.0 g and 587.0 cal/g, respectively. If the fuel contains 5% Hydrogen, calculate its gross and net calorific value. (CO4)
- (b) Write the characteristics of good fuel and discuss classification of fuels with merits and demerits of each class. (CO4)
- (c) Write short notes on the following : (CO4)
- (i) Biogas
- (ii) Solar energy
5. (a) Discuss the principle, classification and applications of fuel cell. (CO5)
- (b) What is electrochemical series ? Discuss applications of electrochemical series. (CO5)

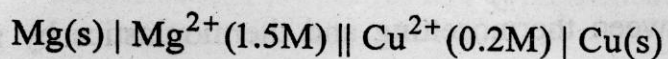
P. T. O.

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- (b) What is electrochemical series ? Discuss applications of electrochemical series. (CO5)

P. T. O.

(c) For the cell :

(CO5)



$$\left[E^\circ_{\text{Mg}^{2+}/\text{Mg}} = -2.37\text{V} \text{ and } E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34 \right]$$

(i) Identify the cathode and the anode.

(ii) Write the electrode reactions.

(iii) Calculate the cell potential.